

Exploration Sprint

For curiosity-driven "what ifs" that aren't research questions yet. This is the precursor to the Research Question Sharpener — use it to find out if there's something worth sharpening.

BEFORE YOU START

The What-If

What are you curious about? State it naturally. "What would happen if..." or "I wonder whether..." is fine here.

Why this, why now?

What prompted this curiosity? A paper you read, an observation, a conversation, a hunch? Capturing the origin helps you notice what kinds of intuitions tend to be productive.

EXPECTATIONS

Without written expectations, you can't recognize surprise — and surprise is what turns exploration into research.

What do you expect to observe?

Be specific. "I expect that scaling cellular automata 100x will produce qualitatively similar dynamics" or "I expect performance to degrade gradually, not suddenly." Vague expectations can't be violated.

Why do you expect this?

What's your reasoning or intuition? This helps you identify which of your assumptions was wrong if the expectation is violated.

What would genuinely surprise you?

Describe 1-2 outcomes that would make you stop and say "wait, that's weird." These are the signals you're looking for during the sprint.

FRUITFULNESS PRE-CHECK

Not everything deserves exploring. Before investing time, check whether the best possible outcome of this sprint would actually lead somewhere worth going.

If this surprises you, then what?

Imagine the most interesting possible outcome. What downstream consequences would it have? What research question would it open? If the best case is "huh, neat" and nothing more — reconsider.

Who would care?

Name a specific community, researcher, or field that would find the result meaningful. If you can't name anyone beyond yourself, the exploration may be intellectually fun but strategically pointless.

Does this connect to our research directions?

Which LossFunk research direction does this fall under? If it doesn't connect to any — is it worth lab time, or is it a personal side interest?

TIME BOX

Sprint duration:	■ 3 days ■ 1 week ■ 2 weeks ■ _____
Check-in date:	____ / ____ / _____
Hard stop date:	____ / ____ / _____

At the check-in: is anything surprising emerging? If not, consider stopping early. At the hard stop: make the decision below. No extensions without a new sprint.

What 'done' looks like

What's the minimum you need to run/build/read to know whether something interesting is here? Keep it small. This is a scout mission, not a campaign.

AFTER THE SPRINT

Fill this in at the end of your time box. Be honest.

What did you actually observe?

Describe what happened. Focus on what matched or didn't match your expectations.

Were your expectations violated?

If yes — how specifically? Which assumption was wrong? If no — that's fine, but it means there's probably not a paper here.

THE DECISION

Pick one. Be disciplined — the whole point of the time box is to force a decision rather than letting exploration drift.

GRADUATE	SHELVE	PIVOT
Surprise found. "I expected X but observed Y. The gap is interesting because:" ----- ----- ----- → Pull out the Research Question Sharpener	No surprise found. Expectations confirmed or results ambiguous. No shame — most explorations end here. File your notes for future reference. What I learned anyway: ----- ----- -----	Something adjacent is interesting. The original what-if wasn't fruitful, but you noticed something else. Start a new sprint for the new what-if. The new what-if: ----- ----- -----

The LossFunk Research Flow

Curiosity ("what if...") → **Exploration Sprint** (this template, time-boxed) → **Surprise found?**

Yes → **Research Question Sharpener** (one-pager) → **Project**

No → **Shelve** or **Pivot** to new sprint (no guilt, this is how it works)